7. Scenario: You are working as a data analyst for an e-commerce company. You have been given a dataset containing information about customer orders, stored in a Pandas DataFrame named order_data. The DataFrame has columns for customer ID, order date, product name, and order quantity. Your task is to analyze the data and answer specific questions about the orders. Question: Using Pandas DataFrame operations, how would you find the following information from the order_data DataFrame:

1. The total number of orders made by each customer.

2. The average order quantity for each product.

3. The earliest and latest order dates in the dataset.

AIM:

To analyze customer orders using Pandas DataFrame operations and find the total orders per customer, average order quantity per product, and earliest and latest order dates.

PROCEDURE:

1. Import the Pandas library.
2. Create a DataFrame containing customer order details.
3. Count orders for each customer using groupby() and count().
4. Calculate average order quantity for each product using mean().
5. Find the earliest and latest order dates using min() and max().
6. Display the results.

PROGRAM CODE:

```
import pandas as pd
order_data = pd.DataFrame({
    'Customer_ID': ['C1', 'C2', 'C1', 'C3', 'C2'],
    'Order_Date': ['2026-01-10', '2026-01-12',
                   '2026-01-15', '2026-01-18', '2026-01-20'],
    'Product_Name': ['Pen', 'Book', 'Book', 'Pen', 'Pen'],
    'Order_Quantity': [2, 3, 4, 5, 1]
})
order_data['Order_Date'] =
pd.to_datetime(order_data['Order_Date'])
print("Total Orders per Customer:")
print(order_data.groupby('Customer_ID').size())
print("\nAverage Quantity per Product:")
print(order_data.groupby('Product_Name')['Order_Quantity'].m
ean())
print("\nEarliest Order Date:", order_data['Order_Date'].min())
print("Latest Order Date:", order_data['Order_Date'].max())
```

INPUT:

- Customer IDs: C1, C2, C1, C3, C2
- Order Dates: 2026-01-10, 2026-01-12, 2026-01-15, 2026-01-18, 2026-01-20
- Product Names: Pen, Book, Book, Pen, Pen
- Order Quantities: 2, 3, 4, 5, 1

OUTPUT:

```
print("Latest Order Date:", order_data['Order_Date'].max())


Total Orders per Customer:
Customer_ID
C1    2
C2    2
C3    1
dtype: int64

Average Quantity per Product:
Product_Name
Book    3.500000
Pen     2.666667
Name: Order_Quantity, dtype: float64

Earliest Order Date: 2026-01-10 00:00:00
Latest Order Date: 2026-01-20 00:00:00
```

RESULT:

Thus, the customer order counts, average product quantities, and earliest and latest order dates were successfully determined using Pandas.